

MATH210B Homework 2

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1 Problem 2

Let's first show that the map taking a set X to $\mathbb{Z}\langle X \rangle$ is left adjoint to the forgetful functor taking a (possibly noncommutative ring R) to its underlying set.

Let $\phi : X \rightarrow S$ be a set map. Define $\phi^* : \mathbb{Z}\langle X \rangle \rightarrow S$ by taking $x \in X$ to $\phi(x)$. Let's show that this is a ring homomorphism.

We'll use the following lemma:

Lemma 1.1. Assume A is a ring that is also a free Abelian group with basis β . Assume $\beta \cdot \beta \subset \beta$. Let $f : A \rightarrow B$ be a group homomorphism such that it's a ring homomorphism when restricted to β . Then, f is a ring homomorphism.

Proof. Let $x, y \in A$. Then, $x = \sum_{i=1}^n a_i v_i$ and $y = \sum_{i=1}^n b_i v_i$ where $a_i, b_i \in \mathbb{Z}$ (possibly zero) and $v_i \in \beta$. By distributivity,

$$xy = \sum_{i=1}^n \sum_{j=1}^n a_i b_j v_i v_j$$

Then,

$$\begin{aligned} f(xy) &= f\left(\sum_{i=1}^n \sum_{j=1}^n a_i b_j v_i v_j\right) \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i b_j f(v_i v_j) \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i b_j f(v_i) f(v_j) \\ &= \sum_{i=1}^n f(a_i v_i) \sum_{j=1}^n f(a_j v_j) \\ &= f(x) f(y) \end{aligned}$$

□

Now, notice that ϕ^* is an Abelian group since it matches the coproduct in the category of Abelian groups, and it satisfies the conditions of the lemma. Thus, it's a ring homomorphism.

The fact that these maps are adjoint is now immediate.

Let $X = R$ as a set. Then, there's a surjective ring homomorphism from $\mathbb{Z}\langle X \rangle$ to R which is given by $r \mapsto r$. Thus, we have the desired result.

2 Problem 3

Let A, B be (possibly noncommutative) rings.

By Problem 2, we know that $A \equiv \mathbb{Z}[X]/I$ and $B \equiv \mathbb{Z}[Y]/J$ for some sets X, Y and ideals I, J of the ring R .

Let the coproduct of A and B be $\mathbb{Z}[X \sqcup Y]/(I + J)$.

We'll show that this is the coproduct by showing that it satisfies the universal property.

Notice that $\mathbb{Z}[X]/I$ injects into $\mathbb{Z}[X \sqcup Y]/(I + J)$ since I gets mapped to $I + J$ where we send $i \mapsto i + 0$. By a similar argument with B , we have injective ring homomorphisms into $\mathbb{Z}[X \sqcup Y]/(I + J)$.

Moreover, given two ring homomorphisms from $\mathbb{Z}[X]$ and $\mathbb{Z}[Y]$, this uniquely defines a ring homomorphism from $\mathbb{Z}[X \sqcup Y]$ using the universal property in Problem 2. Moreover, this ring homomorphism factors through the canonical injections since it acts the same way on the basis. We thus conclude our proof.

3 Problem 5

Let $\mathcal{P}$ be the set of all prime ideals in R .

Order $\mathcal{P}$ by reverse inclusion. To apply Zorn's Lemma, we should show that every chain has a maximal (which in this case means minimal using set inclusion) element.

Let $P_1 \supseteq P_2 \supseteq \dots$ be chain in $\mathcal{P}$. Take the intersection of all the ideals in the chain. This gives another prime ideal contained in all the ideals in the chain. Thus, by Zorn's Lemma, we conclude the proof.